Cumulative hierarchies of universes and their equivalence in dependent type theory

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Abstract

We present a modular and algorithmic proof for the equivalence of different presentations of type theory, notably relating Russell and Tarski presentations for a cumulative hierarchy of universes. A crucial lemma is the injectivity of type formers, which we get by proving normalization for the Tarski-style theory via an adaptation of the gluing technique. Our work can be seen as an extension of initiality results to more informal presentations of type theory.

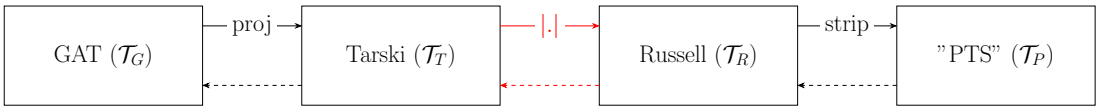
There are two different ways to formulate the notion of universe in type theory. Martin-Löf's first versions [10, 11] were formulated à la Russell, where a universe is seen as a type of types. Later, an alternative version à la Tarski [12], where a universe is seen as a type of codes along with a decoding function, was also introduced. Usually, Russell universes are regarded as an informal notation, as [9] suggested.

When presenting Type Theory as a Generalised Algebraic Theory, Tarski universes are mandatory, and one can show that the Tarski system presents the initial model of such a theory, following the method presented in [13] and formalised in [6]. Hence, it would be essential to have an equivalence result between the two presentations of universes, extending the initiality result to Russell universes. We also relate the generalised algebraic presentation with a more common syntax inherited from pure type systems, without type annotations in λ and app.

Some recent works [2, 14] show the equivalence between these two formulations, but these arguments do not apply to formulations of type theory with equality judgements. In this paper we show the equivalence for this case by a method due to Streicher [13] that seems more modular and transparent than the one in [2, 14]. The work of [2] provides the structural equations necessary for this equivalence, by achieving what he calls *full reflection*.

Therefore, we present a proof of equivalence between Russell and Tarski universes for type theory with dependent products, cumulative universes and judgmental $\beta\eta$ equality. This proof adapts to a theory with universe polymorphism and internal universe levels such as in [3]. Our work relies on basic properties of Type Theory like no confusion and injectivity of type constructors, which seem to be strongly related to conversion [8]. We get those properties by showing normalisation of the Tarski theory, by an adaptation of the gluing proof from [5] Then, we show the proper equivalence result by adapting the method of [13]. The proof is elementary and provides an algorithm to translate from one to the other. We formalised the proof of equivalence entirely in Rocq, taking the results given by normalization as axioms. The development is available on GitHub.

We consider a hierarchy of theories with natural forgetful maps between them. To show their equivalences, we prove that those forgetful maps all have well-defined sections. We will focus on the equivalence highlighted in red : Russel and Tarski universes.



The theory with Tarski universes $\mathcal{T}_T$ has all the structural equations listed by [2] :

$$T_l(t_n^l(X)) = T_n(X) \text{ for } l < n$$

$$T_n(\mathsf{U}_m^n) = \mathsf{U}_m \text{ for } m < n$$

$$t_n^p(\Pi^n\ X\ Y) = \Pi^p\ t_n^p(X)\ t_n^p(Y) \text{ for } n < p$$

$$T_n(\Pi^n\ X\ Y) = \Pi(T_n(X), T_n(Y)) \text{ for } n \geq 0$$

$$t_n^p(t_m^n(X)) = t_m^p(X) \text{ for } m < n < p$$

$$t_n^p(\mathsf{U}_m^n) = \mathsf{U}_m^p \text{ for } m < n < p$$

With injectivity of constructors in $\mathcal{T}_T$, we prove the equivalence of Tarski and Russell universes. For that we consider the following erasure operation, which forms a forgetful map from Tarski universes to Russell ones. It is easy to prove that it preserves derivations.

- $|\mathsf{U}_n| = \mathsf{U}_n$.
- $|\mathsf{U}_m^n| = \mathsf{U}_m$.
- $|\mathsf{app}(A, B, c, a)| = \mathsf{app}(|A|, |B|, |c|, |a|)$.
- $|\lambda(A, B, b)| = \lambda(|A|, |B|, |b|)$.
- $|v_i| = v_i$.
- $|T_n(a)| = |a|$.
- $|t_m^n(a)| = |a|$.
- $|\Pi^n\ a\ b| = \Pi(|a|, |b|)$.
- $|\Pi(A, B)| = \Pi(|A|, |B|)$.

Then, we want to show that this forgetful map has an inverse. For this we need the following result, stating the injectivity of erasure up to judgemental equality.

Uniqueness up to erasure. If $\Gamma \vdash u_0 : A_0$ and $\Gamma \vdash u_1 : A_1$ in $\mathcal{T}_T$, with $|u_0| = |u_1|$ and $|A_0| = |A_1|$, then : $\Gamma \vdash A_0 = A_1$ and $\Gamma \vdash u_0 = u_1 : A_0$.

Finally, we get the following equivalence theorem :

Equivalence. If $\Gamma \vdash J$ in the system $\mathcal{T}_R$, then there exists a unique, up to conversion judgment $\Gamma_1 \vdash J_1$ in the system $\mathcal{T}_T$ such that $|\Gamma_1| = \Gamma$ and $|J_1| = J$.

Extensions. The proof is modular and should adapt to many common variants of dependent type theory. For instance, it adapts easily to a theory with identity types, or to inductive types without structural subtyping (discussed in [9]).

However, when adding structural subtyping, the system might not be that well-behaved as canonicity fails to hold. We would need to add a lot of other rules such as $\mathsf{nil}_A = \mathsf{nil}_B$ for lists, which we believe contradict the spirit of canonical objects. However, there are some cases where structural subtyping could be useful, as described in the paper [7]. The recent work [1] gives a very general framework to recover the right structural equalities.

Universe polymorphism. We consider polymorphic universe levels such as described in [3], and more recently presented as a generalized algebraic theory in [4]. If we assume injectivity of type constructors for this theory, we can adapt our equivalence proof to this system.

References

[1] Arthur Adjedj, Meven Lennon-Bertrand, Thibaut Benjamin, and Kenji Maillard. Adaptt: Functoriality for dependent type casts. *Proc. ACM Program. Lang.*, (POPL), 2026.

[2] Ali Assaf. A calculus of constructions with explicit subtyping. *TYPES*, 2014.

[3] Marc Bezem, Thierry Coquand, Peter Dybjer, and Martín Escardó. Type theory with explicit universe polymorphism. *TYPES*, 2023.

[4] Marc Bezem, Thierry Coquand, Peter Dybjer, and Martín Escardó. A generalized algebraic theory for type theory with explicit universe polymorphism. *Electronic Proceedings in Theoretical Computer Science*, 2026.

[5] Thierry Coquand. Canonicity and normalisation for dependent type theory. *CoRR*, 2019.

[6] Menno de Boer. A proof and formalization of the initiality conjecture of dependent type theory, 2020.

[7] Théo Laurent, Meven Lennon-Bertrand, and Kenji Maillard. Definitional functoriality for dependent (sub)types – extended version. *CoRR*, 2024.

[8] Meven Lennon-Bertrand. What Does It Take to Certify a Conversion Checker? In *FSCD*, 2025.

[9] Zhaohui Luo. Notes on universes in type theory, 2012.

[10] Per Martin-Löf. An intuitionistic theory of types: Predicative part. 1975.

[11] Per Martin-Löf. Constructive mathematics and computer programming. 1982.

[12] Per Martin-Löf. Intuitionistic type theory. *Bibliopolis*, 1984.

[13] Thomas Streicher. *Semantics of Type Theory: Correctness, Completeness and Independence Results*. 2012.

[14] François Thiré. *Interoperability between proof systems using the logical framework Dedukti*. PhD thesis, 2020.